

ROLL NO:250701573

EXPERIMENT NO: 11 EXPERIMENT TO UNDERSTAND KNN ALGORITHM FOR A  
GIVEN DATA SET

PROGRAM:

```
import numpy as np
import pandas as pd

df = pd.read_csv('Iris_modified.csv')
df
df.info()
df.variety.value_counts()
df.head()

features = df.iloc[:, :-1].values
label = df.iloc[:, 4].values

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

xtrain, xtest, ytrain, ytest = train_test_split(
    features, label, test_size=0.2, random_state=42
)

print(model_KNN.score(xtrain, ytrain))
print(model_KNN.score(xtest, ytest))

from sklearn.metrics import confusion_matrix

confusion_matrix(label, model_KNN.predict(features))

from sklearn.metrics import classification_report
```

```
print(classification_report(label, model_KNN.predict(features)))
```

OUTPUT:

| sepal.length | sepal.width | petal.length | petal.width | variety |           |
|--------------|-------------|--------------|-------------|---------|-----------|
| 0            | 5.2         | 3.5          | 1.5         | 0.3     | Setosa    |
| 1            | 4.7         | 3.2          | 1.5         | 0.3     | Setosa    |
| 2            | 4.6         | 3.2          | 1.2         | 0.4     | Setosa    |
| 3            | 4.7         | 3.2          | 1.5         | 0.1     | Setosa    |
| 4            | 5.0         | 3.4          | 1.5         | 0.3     | Setosa    |
| ...          | ...         | ...          | ...         | ...     | ...       |
| 145          | 6.7         | 3.0          | 5.0         | 2.3     | Virginica |
| 146          | 6.5         | 2.6          | 4.9         | 1.8     | Virginica |
| 147          | 6.4         | 2.8          | 5.4         | 2.0     | Virginica |
| 148          | 6.1         | 3.4          | 5.2         | 2.2     | Virginica |
| 149          | 6.0         | 3.1          | 5.3         | 1.8     | Virginica |

150 rows × 5 columns

```
<class 'pandas.DataFrame'>
```

```
RangeIndex: 150 entries, 0 to 149
```

```
Data columns (total 5 columns):
```

| # | Column       | Non-Null Count | Dtype   |
|---|--------------|----------------|---------|
| 0 | sepal.length | 150 non-null   | float64 |
| 1 | sepal.width  | 150 non-null   | float64 |
| 2 | petal.length | 150 non-null   | float64 |
| 3 | petal.width  | 150 non-null   | float64 |
| 4 | variety      | 150 non-null   | str     |

```
dtypes: float64(4), str(1)
```

```
memory usage: 6.0 KB
```

---

```
variety
```

```
Setosa      50
```

```
Versicolor  50
```

```
Virginica   50
```

```
Name: count, dtype: int64
```

| sepal.length | sepal.width | petal.length | petal.width | variety |        |
|--------------|-------------|--------------|-------------|---------|--------|
| 0            | 5.2         | 3.5          | 1.5         | 0.3     | Setosa |
| 1            | 4.7         | 3.2          | 1.5         | 0.3     | Setosa |
| 2            | 4.6         | 3.2          | 1.2         | 0.4     | Setosa |
| 3            | 4.7         | 3.2          | 1.5         | 0.1     | Setosa |
| 4            | 5.0         | 3.4          | 1.5         | 0.3     | Setosa |

0.9416666666666667

0.9666666666666667

```
array([[50,  0,  0],
       [ 0, 48,  2],
       [ 0,  6, 44]])
```

| precision    | recall | f1-score | support |     |
|--------------|--------|----------|---------|-----|
| Setosa       | 1.00   | 1.00     | 1.00    | 50  |
| Versicolor   | 0.89   | 0.96     | 0.92    | 50  |
| Virginica    | 0.96   | 0.88     | 0.92    | 50  |
| accuracy     |        |          | 0.95    | 150 |
| macro avg    | 0.95   | 0.95     | 0.95    | 150 |
| weighted avg | 0.95   | 0.95     | 0.95    | 150 |